Lending Club Loan Default Prediction Project

This document explains the complete workflow implemented in the Jupyter notebook for the Lending Club Loan Default Prediction project. The notebook focuses on building a machine learning pipeline capable of predicting whether a borrower is likely to default on a loan. The project uses the Lending Club dataset and applies preprocessing, feature engineering, model training, and model persistence techniques.

1. Dataset Acquisition

The notebook begins by downloading and loading the Lending Club dataset using KaggleHub. Two datasets are imported:

• Accepted loans dataset
• Rejected loans dataset

The accepted loans dataset is primarily used for training the predictive model because it contains the actual loan outcomes.

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2. Data Loading and Inspection

The notebook loads the CSV files using pandas with the parameter **low_memory=False** to avoid datatype warnings. After loading, the notebook extracts and reviews all dataset columns to understand the available features.

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3. Feature Selection and Column Removal

A major preprocessing step involves removing unnecessary or potentially harmful columns. The notebook removes:

• Identifier columns
• Administrative fields
• Text-heavy columns
• Post-loan information that could cause data leakage
• Columns with little predictive value

This step is extremely important because financial prediction systems must avoid leakage from future information.

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4. Data Cleaning

The notebook handles missing values and invalid data entries. Some columns with excessive missing values are removed, while other preprocessing operations ensure the dataset becomes suitable for machine learning.

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5. Loan Status Processing

The notebook filters loan statuses to keep only finalized loan outcomes. Ongoing loans such as:

• Current
• In Grace Period

are removed because their final repayment behavior is still unknown.

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6. Feature Engineering

Multiple features are transformed into machine-learning-friendly formats.

Examples include:

• Encoding employment length
• Converting loan term into numeric format
• Processing credit history dates
• Transforming categorical variables

The notebook also creates useful numerical representations from raw financial information.

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7. Target Variable Creation

A binary target variable is created to represent loan default behavior.

Typically:

• 1 = Default / Charged Off
• 0 = Fully Paid

This converts the problem into a binary classification task.

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8. Handling Imbalanced Data

Loan default datasets are naturally imbalanced because most borrowers repay successfully. The notebook calculates class imbalance and uses **scale_pos_weight** with XGBoost to improve learning performance on minority default cases.

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9. Train-Test Split

The dataset is divided into training and testing sets. This ensures the model is evaluated on unseen data and reduces the risk of overfitting.

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10. Model Training

The notebook trains an XGBoost classifier, which is one of the most effective algorithms for structured financial datasets.

The model learns patterns related to:

• Debt-to-income ratio
• Employment length
• Loan amount
• Credit history
• Verification status

and many other financial indicators.

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11. Model Evaluation

The notebook evaluates the trained model using classification metrics and probability predictions. This helps determine how accurately the system predicts loan default risk.

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12. Model Persistence

The final trained XGBoost model is saved using Python pickle files. Feature names are also stored to maintain consistent feature ordering during deployment and inference.

Saved files include:

• xgb_model.pkl
• feature_names.pkl

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13. Project Significance

This project demonstrates a practical financial risk prediction system using machine learning. It can serve as the foundation for:

• AI-powered loan approval systems
• Credit risk analysis platforms
• FinTech applications
• Explainable AI financial assistants

The project is suitable as a prototype-level Final Year Project because it combines data preprocessing, machine learning, and deployment readiness.

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14. Future Improvements

Several enhancements can further improve the project:

• Hyperparameter tuning
• SHAP explainability integration
• Cross-validation
• Model comparison with Random Forest and Logistic Regression
• FastAPI deployment
• Integration with locally hosted LLMs using Ollama and Llama 3

Project Summary

Component	**Description**
Dataset	Lending Club Accepted Loans Dataset
Problem Type	Binary Classification
Algorithm	XGBoost Classifier
Target	Loan Default Prediction
Saved Model	xgb_model.pkl
Features	51 engineered features
Deployment Goal	AI-powered loan risk assistant

The notebook successfully demonstrates the complete lifecycle of a machine learning project for financial risk prediction. It includes data preparation, preprocessing, feature engineering, model training, evaluation, and deployment preparation. The project forms a strong foundation for integrating machine learning with modern AI systems such as locally hosted large language models.